

Functional genetic report · longevity

Aorta and Aneurysms

Sérgio F · Report date: 02-10-2026

The aorta and valves age by elastin fragmentation, collagen stiffness, inflammation and calcification. This report reads the genes of the abdominal aorta aneurysm, the diameter of the thoracic aorta, the calcification of the aortic valve, the mitral prolapse and cerebral aneurysms. Only genes (chip + local imputation + polygenic scores).

Contents

• Summary	3
• How to read this report	4
• Polygenic scores	5
• Abdominal aneurysm	6
• Diameter of the thoracic aorta and wall (fibrillin, elastin, collagen)	7
• Aortic valve: calcification and stenosis (LPA, PALMD, TEX41, IL6)	8
• Mitral valve: prolapse and regurgitation	9
• Intracranial aneurysms and arterial dissections	10
• Mechanisms of ageing	11
• Practical suggestions	12
• Limitations	13
• Methods and sources	14

Summary

26

SNPs analyzed

20

in the chip

6

imputed (DR2 $\geq$ 0.8)

0

SNP not found in the data

9

Polygenic scores

Abdominal aneurysm is mainly a disease of wall aging (inflammation, metalloproteases, loss of smooth muscle cells) and genetic sharing with atherosclerosis (9p21, LDL, Lp(a), IL-6). The aneurysm of the thoracic aorta depends more on the matrix: fibrillin-1 (FBN1), elastin (ELN), lysyl oxidase (LOX) and TGF- β signal. Calcified aortic stenosis is the most frequent valvular disease after 65 years and has as main genetic factors Lp(a), palmelfin (PALMD), TEX41 and IL-6. Mitral prolapse is the main cause of primary mitral regurgitation.

In summary: abdominal aorta with mean genetic risk (1.1 M score variants at the 42nd percentile), aided by light protective factors (IL6R Ala358, LDLR, DAB2IP without risk, Lp(a) low genetic) against risk homozygous 9p21. The unfavourable point is the genetically wider ascending thoracic aorta: diameter score at the 79th percentile, with a copy of the major aorta alleles in two FBN1 signals and in the ULK4. In the aortic valve, Lp(a) and IL-6 are favourable and TEX41 is unfavourable (aortic stenosis score at the 45th percentile in this score; the valves report scores give p75 to p95,1, so the reference reading is p89-95). Intracranial aneurysms with low genetic risk and mitral valve with medium risk.

Favourable

- **Low genetic Lp(a)** (LPA rs10455872 AA and rs3798220 TT) and **IL6 -597 GG**: without the two major genetic factors of calcified aortic stenosis.
- **Abdominal aneurysm protectors**: IL6R Asp358Ala (one copy Ala), LDLR rs6511720 GT (one copy of the low LDL allele), DAB2IP rs7025486 GG and HDAC9 GG; compact score of abdominal aorta aneurysm at the 26th percentile.
- **Mitral valve and cerebral aneurysms**: LMCD1 AA and TNS1 GG (without the mitral prolapse alleles of the two strongest loci; global score at the 55th percentile); RBBP8 AA and intracranial aneurysm score at the 10th percentile; spontaneous coronary dissection score at the 13th percentile.

Unfavourable

- **Genetically wider ascending aorta**: diameter score (1.1 M variants) at the 79th percentile; FBN1 rs2118181 TC and rs1036477 AG (each less common allele increases the diameter) and ULK4 rs35475272 TC.
- **9p21 (CDKN2B-AS1) rs1333049 CC**: homozygous for the risk allele of abdominal aortic aneurysm and coronary artery disease.
- **TEX41 rs2252641 CC**: homozygous for the allele of calcified aortic stenosis (OR $\sim$ 1.1 by allele); PALMD TG.

Interpretation: What counts most here is the ascending thoracic aorta with a genetic tendency to be wider, a trait that only one image confirms. Telmisartan is the most rational drug to protect the aorta wall. Avoid fluoroquinolones when there is an alternative.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 20 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

Two abdominal aneurysm scores (the compact, the 26th percentile, and one of 1.1 M variants, the 42nd percentile), two of ascending aorta diameter (89 variants, the 36th percentile) were used. 1.1 M variants, the 79th percentile), one of aortic stenosis (1.1 M variants, the 45th percentile), one of aortic valve disease (36 variants, the 48th percentile) and one of mitral prolapse (1.1 M variants, the 55th percentile). Intracranial aneurysm and spontaneous coronary dissection scores use GWAS Catalog loci (excluding studies from other populations). The GWAS Catalog score for the descending aorta was excluded because it mixed different units.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Abdominal aneurysm (1.1 M variants)	PGS003972	1 085 781 of 1 118 997 (97%)	p42	Typical / no judgement
Abdominal aneurysm (compact score)	PGS000753	27 of 29 (93%)	p26	Typical / no judgement
Diameter of ascending aorta (1.1 M variants)	PGS004927	1 080 959 of 1 111 632 (97%)	p79	Typical / no judgement
Diameter of ascending aorta (89 variants)	PGS002267	82 of 89 (92%)	p36	Typical / no judgement
Aortic stenosis (1.1 M variants)	PGS005252	1 086 572 of 1 119 377 (97%)	p45	Typical / no judgement
Prolapse of mitral valve (1.1 M variants)	PGS002271	1 066 402 of 1 097 364 (97%)	p55	Typical / no judgement
Non-rheumatic aortic valve disease (36 variants)	PGS001025	26 of 36 (72%)	p48	Typical / no judgement
Intracranial aneurysm (GWAS Catalog score)	C_ANEUR_IC	10 of 10 (100%)	p10	Favourable
Spontaneous coronary artery dissection (3 loci)	C_SCAD	3 of 3 (100%)	p13	Favourable

Abdominal aneurysm

The aneurysm of the abdominal aorta (diameter ≥ 3 cm) is about 4–6 times more frequent in men and increases after 65 years. Genetic factors are shared with atherosclerosis (9p21, LDL, Lp(a)) and with wall inflammation (IL-6R, DAB2IP).

Polygenic scores of this theme: Abdominal aneurysm (1.1 M variants) (PGS003972): the 42nd percentile

Typical / no judgement; Abdominal aneurysm (compact score) (PGS000753): the 26th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CDKN2B-AS1 9p21	rs1333049	CC	chip	C	2	0.48	Locus 9p21 (ANRIL/CDKN2B): C allele increases coronary disease and abdominal aortic aneurysm; it acts on the senescence and proliferation of smooth muscle cells. (strong evidence)	Unfavourable
DAB2IP	rs7025486	GG	chip	A	0	0.22	DAB2IP: A allele increases abdominal aortic aneurysm and infarction. (strong evidence)	Favourable
DAB2IP 2 signal	rs10985343	TT	imputation DR2=0.97	C	0	0.24	Second sign of DAB2IP in the biobank: T allele protects, C increases risk. (moderate evidence)	Typical / no judgement
ADAMTS8	rs4936098	AG	imputation DR2=0.99	G	1	0.65	Matrix metalloprotease: allele A reduces the risk of aneurysm, G (common) is the risk. (moderate evidence)	Typical / no judgement
IL6R Asp358Ala	rs2228145	AC	chip	C	1	0.36	The Ala358 form (C allele) reduces the IL-6 signal in the membrane and protects from abdominal aortic aneurysm and coronary disease. (strong evidence)	Favourable
LDLR	rs6511720	GT	chip	T	1	0.11	The T allele (less common) lowers LDL and reduces the risk of aneurysm and coronary disease. (strong evidence)	Favourable
SORT1	rs599839	GA	chip	A	1	0.78	SORT1/CELSR2 (1p13): A allele (common) increases LDL, aortic aneurysm and aortic stenosis. (moderate evidence)	Typical / no judgement

Favourable

- IL6R Ala358 (1 copy)
- LDLR rs6511720 GT (1 protective allele)
- DAB2IP rs7025486 GG
- Compact aneurysm score at the 26th percentile

Unfavourable

- 9p21 rs1333049 CC
- SORT1 rs599839 GA and ADAMTS8 AG (typical)

Interpretation: Medium-low profile: 9p21 homozygous risk is compensated by IL6R Ala358, LDLR protective and DAB2IP without risk. The 1.1 M variant score is at the 42nd percentile. The main modifiable factors of abdominal aneurysm are tobacco (the largest of all), LDL and inflammation.

Diameter of the thoracic aorta and wall (fibrillin, elastin, collagen)

The ascending aorta diameter has a strong genetic component (GWAS of cardiac images with more than 30 000 people). The rare variants of FBN1 cause Marfan syndrome; the common ones, read here, change the diameter in fraction of millimeters per allele.

Polygenic scores of this theme: Diameter of ascending aorta (1.1 M variants) (PGS004927): the 79th percentile **Typical / no judgement** ; Diameter of ascending aorta (89 variants) (PGS002267): the 36th percentile **Typical / no judgement**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
FBN1	rs2118181	TC	chip	C	1	0.11	Fibrillin-1 (the Marfan syndrome protein): the C allele, less common, increases the ascending thoracic aorta diameter. (moderate evidence)	Unfavourable
FBN1 2 signal	rs1036477	AG	chip	G	1	0.11	Second common sign of FBN1: G allele increases the ascending and descending aorta area. (moderate evidence)	Unfavourable
ELN	rs6974735	AG	imputation DR2=0.99	G	1	0.48	Elastin: the G allele increases the ascending aorta area. (moderate evidence)	Typical / no judgement
ULK4	rs35475272	TC	imputation DR2=1.00	C	1	0.19	ULK4: the C allele, less common, increases the ascending aorta area; also aortic dissection locus. (moderate evidence)	Unfavourable
HDAC9	rs2107595	GG	chip	A	0	0.16	HDAC9: Allele A increases the diameter of the descending aorta, the calcification of the abdominal aorta, the large artery stroke and coronary disease. (strong evidence)	Favourable
LOX Arg158Gln	rs1800449	CC	chip	T	0	0.16	Lysyl oxidase, which binds collagen and elastin; T allele reduces activity and is associated with coronary disease (and, in families, with thoracic aneurysm). (moderate evidence)	Typical / no judgement

Favourable

- HDAC9 rs2107595 GG (without the major descending aorta allele and calcification)
- LOX Arg158 (CC, without the lowest activity variant)
- 89 variant score at the 36th percentile

Unfavourable

- 1.1 M diameter score variants at the 79th percentile
- FBN1 rs2118181 TC and rs1036477 AG
- ULK4 rs35475272 TC
- ELN rs6974735 AG (typical)

Interpretation: The two scores differ (89 variants at the 36th percentile; 1.1 M at the 79th percentile; the highest score, consistent with FBN1 and ULK4, is the most informative. Genetic trend for a slightly wider ascending aorta, without any rare variant of Marfan read (the chip does not read the rare variants of FBN1). The AT1 blockers reduced the progression of aortic root dilation in Marfan syndrome trials, and telmisartan is already in the plan.

Aortic valve: calcification and stenosis (LPA, PALMD, TEX41, IL6)

Calcified aortic stenosis begins with thickening and calcification of the leaflets (aortic sclerosis) and progresses slowly for decades. Lp(a) is the strongest causal factor; PALMD, TEX41 and IL-6 are the other main loci.

Polygenic scores of this theme: Aortic stenosis (1.1 variants) (PGS005252): the 45th percentile

Typical / no judgement; Non-rheumatic aortic valve disease (36 variants) (PGS001025): the 48th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
LPA	rs10455872	AA	chip	G	0	0.07	Lipoprotein(a): G allele greatly increases Lp(a), calcified aortic stenosis and coronary disease. (strong evidence)	Favourable
LPA I4399M	rs3798220	TT	chip	C	0	0.01	According to high Lp(a) allele (C, rare). (strong evidence)	Favourable
PALMD	rs6702619	TG	chip	G	1	0.47	PalmDelphine: locus of calcified aortic stenosis; the G allele increases the risk of aortic valve disease and the size of the aorta root. (strong evidence)	Typical / no judgement
TEX41	rs2252641	CC	chip	C	2	0.47	TEX41 (2q22): one of the strongest loci of calcified aortic stenosis (OR ~1.1 by C allele). (strong evidence)	Unfavourable
IL6 promoter (-597)	rs1800797	GG	chip	A	0	0.42	IL-6 promoter (in connection with -174G>C): A allele increases aortic stenosis and valve flow velocity; it binds inflammation to valvular calcification. (strong evidence)	Favourable

Favourable

- LPA rs10455872 AA and rs3798220 TT (low genetic Lp(a))
- IL6 -597 GG (without stenosis allele A)
- Aortic valve disease score (36 variants) at the 48th percentile

Unfavourable

- TEX41 rs2252641 CC (2 risk alleles)
- PALMD rs6702619 TG

Interpretation: The two strongest factors (Lp(a) and IL-6) are favourable; the TEX41 homozygous and the PALMD heterozygous pull in the opposite direction. The 1.1 M aortic stenosis score is at the 45th percentile. Low LDL and low ApoB (SORT1 GA, LDLR protector) also reduce valvular calcification.

Mitral valve: prolapse and regurgitation

The mitral valve prolapse affects 2–3% of people and is the main cause of primary (degenerative) mitral regurgitation. The common loci are in genes of development and matrix of leaflets (LMCD1, TNS1) and cardiac muscle (ALPK3, BAG3).

Polygenic scores of this theme: Prolapse of mitral valve (1.1 M variants) (PGS002271): the 55th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
LMCD1	rs171408	AA	chip	G	0	0.21	LMCD1: mitral valve prolapse locus (OR ~1.3 by G allele), the main cause of primary mitral regurgitation. (strong evidence)	Favourable
TNS1	rs12465515	GG	chip	C	0	0.33	Tensin-1 (TNS1/TESHL region): the C allele increases the mitral prolapse; the tensin binds the matrix to the cytoskeleton of the valve leaflets. (strong evidence)	Favourable
ALPK3	rs35828350	GA	imputation DR2=0.99	A	1	0.27	ZNF592/ALPK3: allele A increases mitral prolapse. (moderate evidence)	Typical / no judgement
BAG3	rs17099139	CG	imputation DR2=0.99	C	1	0.73	BAG3 (cochaperone of cardiac muscle autophagy): the C allele, common, increases mitral prolapse. (moderate evidence)	Typical / no judgement

Favourable - LMCD1 rs171408 AA (without G allele, OR ~1.3) - TNS1 rs12465515 GG (without C allele)	Unfavourable - ALPK3 rs35828350 GA - BAG3 rs17099139 CG (typical)
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Interpretation: Genetic risk of middle mitral prolapse, with the two strongest loci without risk alleles (1.1 M score variants at the 55th percentile). Secondary mitral regurgitation, which appears with ventricular dilation, enters the heart failure report.

Intracranial aneurysms and arterial dissections

Intracranial aneurysms (sacular) exist in about 3% of adults and most of them never break; the common genetics (SOX17, RBBP8, 9p21, EDNRA) explains part of the risk. Spontaneous arterial dissections (coronary and cervical) share loci with migraine and fibromuscular dysplasia (PHACTR1, LRP1).

Polygenic scores of this theme: Intracranial aneurysm (GWAS Catalog score) (C_ANEUR_IC): the 10th percentile **Favourable**; Spontaneous coronary artery dissection (3 loci) (C_SCAD): the 13th percentile **Favourable**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
SOX17	rs9298506	AA	chip	A	2	0.83	SOX17 (development of the arterial endothelium): the allele A (common) increases the intracranial aneurysm. (strong evidence)	Typical / no judgement
RBBP8	rs11661542	AA	chip	C	0	0.49	RBBP8 (18q11): C allele increases the intracranial aneurysm. (strong evidence)	Favourable
PHACTR1	rs9349379	AA	chip	A	2	0.61	PHACTR1/EDN1: allele A increases spontaneous coronary artery dissection, fibromuscular dysplasia and migraine, but protects from coronary disease (G is the coronary risk). (strong evidence)	Typical / no judgement
LRP1	rs11172113	TC	chip	T	1	0.62	LRP1: T allele increases spontaneous coronary dissection, aneurysms and migraine. (moderate evidence)	Typical / no judgement

Favourable

- RBBP8 rs11661542 AA (without aneurysm C allele)
- Intracranial aneurysm score at the 10th percentile
- Spontaneous coronary dissection score at the 13th percentile

Unfavourable

- SOX17 rs9298506 AA (common risk allele, ~83%)
- LRP1 rs11172113 TC

Interpretation: Low genetic risk of intracranial aneurysm and dissection. PHACTR1 AA is the genotype that increases coronary dissection and migraine but protects from coronary disease (information). Tobacco is the strongest modifiable factor for the formation and rupture of cerebral aneurysms.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Fragmentation of Elastin	FBN1 (2 signals) and ELN heterozygous	LOX Arg158	Unfavourable in the ascending aorta.
Remodelling of the extracellular matrix	ULK4 TC; ADAMTS8 AG	LMCD1 AA; TNS1 GG; mitral prolapse p55	Mixed.
Stiffness of the Extracellular Matrix	TEX41 CC; PALMD TG	Low Lp(a); IL6 GG	Mixed in aortic valve.
Cell Senescence	9p21 CC	—	Unfavourable.
Chronic Inflammation	—	IL6R Ala358; IL6 -597 GG; DAB2IP GG	Favourable.
Vascular Aging	9p21 CC	LDLR protector; HDAC9 GG; intracranial aneurysm p10	Globally favourable.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Base image of the aorta	Do once an echocardiogram that measures the root and ascending aorta and see the aortic and mitral valves; if normal, repeat only if the doctor indicates it. <i>suggestion (not applied)</i>	P79 diameter score; FBN1, ULK4
Abdominal aneurysm screening	Abdominal ultrasound of the aorta from 65 years (recommended screening in men), or before if there is a family history. <i>suggestion (not applied)</i>	9p21 CC
Fluoroquinolones	Avoid ciprofloxacin, levofloxacin and moxifloxacin when there is an alternative: in observational studies, the risk of aneurysms and dissection of the aorta increases about 1.5-2 and they damage collagen. Tell the doctors. <i>suggestion (not applied)</i>	Broader ascending aorta; FBN1
Telmisartan	Maintaining as prescribed: AT1 blockade reduces the TGF- β signal and metalloproteases of the aorta wall. <i>suggestion (not applied)</i>	FBN1; ULK4
Maximum effort with apnea	If the image shows increased aorta, avoid maximum surveys with prolonged Valsalva maneuver (prefer more repetitions with moderate load) and talk to the cardiologist. <i>suggestion (not applied)</i>	P79 diameter score
Lp(a), LDL and valves	Measure Lp(a) once (if not yet done) to confirm the low value; LDL and ApoB as low as possible for the aortic valve and aorta. <i>suggestion (not applied)</i>	LPA without risk alleles; TEX41 CC
Coordination with the plan	No tobacco and no alcohol. D3 10 000 IU and omega-3 3.2-4 g/day (already taking fish oil). No telomerase activators. Curcumin, luteolin, apigenin and quercetin in pause while ALT is high. <i>suggestion (not applied)</i>	V2 plan rules

Limitations

- The rare variants that cause hereditary aortopathies (FBN1, TGFBR1/2, SMAD3, ACTA2, MYH11, COL3A1) are not read reliably by the chip.
- The aorta diameter scores measure a genetic trend of fraction of millimeter; only the image gives the real diameter.
- The percentiles compare with the reference population (n=633).
- Only genes: does not use image or analysis.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population ($n=633$); effect alleles confirmed by the expected frequency.
- Scores: PGS003972 (abdominal aorta aneurysm), PGS004927 and PGS002267 (aortic ascending diameter), PGS005252 (aortic stenosis), PGS002271 (prolapse) and PGS001025 (aortic valve disease) calculated in this report PGS000753 reused from the cardiovascular report; simple scores of GWAS Catalog (local copy) for intracranial aneurysm and spontaneous coronary dissection.
- Main references: Jones 2017 and Roychowdhury 2023 (GWAS of abdominal aortic aneurysm); Pirruccello 2022 (thoracic aorta diameter); Thanassoulis 2013 (Lp(a) and aortic stenosis); Thériault 2018 (PALMD and TEX41); Dina 2015 and Roselli 2022 (prolapse mitral); Bakker 2020 (intracranial aneurysm); Adlam 2023 (spontaneous coronary dissection); Lacro 2014 (AT1 block in Marfan); Pasternak 2018 (fluoroquinolones and aortic aneurysm).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).